# NewsChain

*A news app that starts from the outcome and rewinds time to reveal the full story.*

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## Contribution

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# 1. App Idea for Prototype

## 1.1 Description of the App

NewsChain is a narrative-driven mobile news application that helps users understand current events by starting from the present outcome and rewinding through past events to explain how and why the news happened. Rather than presenting an isolated headline, NewsChain presents news as a connected chain of events — each node in the chain is a distinct card showing a headline, source, date, and brief 2–3 sentence summary. Users can tap backwards through the chain to trace cause-and-effect relationships from today's outcome all the way to the origin event.

## 1.2 How Is It Novel?

Unlike traditional news apps (Apple News, Google News, BBC News) that prioritize the latest headlines in a chronological feed, NewsChain reverses the timeline of news consumption. By beginning with the outcome and guiding users backwards through relevant context, the app promotes deeper understanding and reduces surface-level, reactionary news reading. No existing mainstream news application presents news as a visual reverse-timeline chain of interconnected events with source attribution at every node.

## 1.3 Transformative Experience for the User

NewsChain creates four distinct shifts in the user experience:

- Grounded Perspective — Users see the present event as a single, clear moment of significance rather than an isolated headline.
- Sense of Discovery — Users experience curiosity and engagement as they uncover past events that led to today's news.
- Narrative-Driven Reading — News consumption becomes story-like, with cause and effect made explicit.
- Critical Thinking — By seeing full event chains, users develop a well-informed understanding of complex events and can better detect misinformation.

App Motto: "Starting with the outcome, trace back to the cause."

# 2. Background Research

## 2.1 Market Competitor Apps

The following apps represent the primary competitive landscape for NewsChain:

- Apple News — Aggregates personalized news in a clean feed; lacks historical story context or timeline navigation.
- Google News — AI-powered news aggregator organized by topic; stories appear in isolation without a reverse-timeline chain feature.
- BBC News App — Editorial news with high credibility; presents chronological article lists without cause-and-effect mapping.
- The New York Times App — Premium content with deep journalism; subscription-based, no visual story chain navigation.

## 2.2 Market Inspiration Apps

The following apps provided design inspiration and conceptual direction for NewsChain:

- Wikipedia — Demonstrates how complex events can be understood through linked historical context and references. Inspired the "chain" linking model in NewsChain.
- Notion / Knowledge Graph Tools — Show how information can be presented as connected nodes rather than linear feeds. Inspired the node-based card design in NewsChain.

# 3. PACT Framework

## 3.1 People

The primary users of NewsChain are:

- University and college students who need contextual understanding of current events for academic or civic engagement.
- Young professionals (ages 25–40) who follow industry news but have limited time to read multiple full articles.
- Casual news readers who feel overwhelmed or under-informed by conventional news feeds.
- Skeptical readers who want to verify source credibility before trusting and sharing a news story.

## 3.2 Activity

Users open NewsChain when they encounter a headline they want to understand more deeply. The core activity is: (1) viewing a trending news card on the home feed, (2) tapping into the story chain starting from the most recent outcome node, and (3) swiping or tapping backwards through historical event nodes to trace the full story. Secondary activities include bookmarking chains, sharing chains, and exploring topics via the search screen.

## 3.3 Context

The app is primarily used during commutes, short breaks, and study sessions. Users are often in mobile, distracted, or time-limited environments — on the SkyTrain, between classes, or on a lunch break. This context demands that content be short, scannable, and easy to navigate with one hand. The reverse-timeline chain format is well-suited to this context because each node delivers digestible value in under 30 seconds.

## 3.4 Technology

NewsChain is designed as a native mobile application (iOS and Android). Key technologies required include:

- Timeline-based navigation — swipe and tap gesture support for navigating event chains.
- Source aggregation — integration with news APIs to compile multi-source event nodes.
- Data visualization — visual chain rendering to connect event nodes.
- Personalization engine — topic preference matching to curate the home feed.
- Bookmarking and offline storage — for saving chains across sessions.

## 4. Value Proposition

NewsChain provides a new kind of reading experience for users who want to understand current events deeply, not just skim headlines. Compared to traditional news apps like Apple News and Google News, NewsChain starts with the outcome of an event and rewinds through past events to reveal cause-and-effect relationships explicitly.

This approach delivers three core value advantages:

- Depth without effort — Users gain full story context in a single, navigable chain rather than searching across five separate articles.
- Trust and transparency — Source attribution and credibility indicators on every node help users assess reliability at a glance.
- Engagement and discovery — The reverse-timeline format turns passive headline-skimming into active story exploration, fostering critical thinking and reducing misinformation susceptibility.

By combining timeline-based navigation, concise node summaries, and visualized event connections, NewsChain makes news understandable and easily consumable within the time constraints of modern life.

# 5. Requirements Gathering (User Study 1)

## 5.1 Purpose and Objectives

The objective of User Study 1 was to understand how people currently consume news and to identify pain points that NewsChain can address. Specifically, the study aimed to:

- Understand the news consumption habits and preferred platforms of the target users.
- Identify how often users feel confused or lacking context when encountering a breaking news headline.
- Determine user interest in a reverse-timeline news format that starts with a headline outcome and progressively reveals background context.
- Assess user preferences for news depth, from brief summaries to full historical chains of events.
- Understand what motivates or discourages users from reading full news articles to discover design priorities.
- Evaluate whether users would want personalization features such as topic preferences and saved story chains.

## 5.2 Choice of User Study and User Sample

User Study Type: An online survey was selected as the user study method. A survey allows for rapid, scalable data collection across a diverse group and is well-suited for gathering both quantitative data (frequency, ratings) and qualitative data (open-ended responses) using Google Forms.

Target Users: The target users for NewsChain are adults across a wide age range who regularly consume digital news, particularly through smartphones. This includes university and college students, young professionals, and casual news readers.

Dual Perspective: A single user type (general digital news readers) was used for this study, as NewsChain is primarily a consumer-facing news application without a content-publisher side.

Sample Size: 12 participants were surveyed. Participants included housemates, friends, and relatives.

## 5.3 Draft of the User Study Survey

### Section A – Demographic Information

- Q1. What is your age range? (Under 18 / 18–24 / 25–34 / 35–44 / 45 or older)
- Q2. What is your occupation? (Student / Working Professional / Freelancer / Unemployed / Other)
- Q3. How many hours per day do you typically spend consuming news? (Less than 30 min / 30 min–1 hour / 1–2 hours / More than 2 hours)

### Section B – Current News Habits

- Q4. Which platform(s) do you use most to follow news? (Social Media / News Apps / News

Websites / TV/Radio / Word of mouth / Other)

- Q5. On a scale of 1–5, how often do you feel you lack background context when you encounter a breaking news headline? (1 = Never, 5 = Always)
- Q6. When you encounter a confusing news story, what do you typically do? (Search online / Ask someone / Ignore it / Skim multiple headlines / Other)

**Section C – Interest in NewsChain's Concept**

- Q7. How interested are you in a news format where stories start from the latest headline and you can tap backwards to see how the story developed over time? (1 = Not interested, 5 = Very interested)
- Q8. How important is it that a news app gives you the full historical context of a story? (1 = Not important, 5 = Very important)
- Q9. Would you prefer news stories presented as: Reverse timeline / Traditional chronological order / No preference
- Q10. On a scale of 1–5, how valuable would it be to see a visual chain connecting past events to a current headline? (1 = Not valuable, 5 = Very valuable)

**Section D – Feature Preferences**

- Q11. Which features would you most want in a news app? (Bookmarking / Personalized feed / Offline reading / Story chain visualization / Share story chain / Dark mode / Audio summary)
- Q12. How important is it for the app to be available offline? (1 = Not important, 5 = Essential)

## 5.4 Data Visualization and Analysis

**Q3 – Daily News Consumption Time (n = 12)**

| Time Spent | Responses | Percentage |
|---|---|---|
| Less than 30 min | 3 | 25% |
| 30 min – 1 hour | 6 | 50% |
| 1 – 2 hours | 2 | 17% |
| More than 2 hours | 1 | 8% |

The majority of users (50%) spend 30 minutes to 1 hour per day on news consumption, indicating a moderate but consistent engagement level. This reinforces the need for concise, scannable content within NewsChain.

**Q5 – Frequency of Lacking Context (Mean Score, n = 12)**

| Metric | Score (out of 5) |
|---|---|
| Mean | 3.8 |
| Median | 4 |

| | |
|---|---|
| Mode | 4 |

A mean score of 3.8 out of 5 indicates that users frequently feel they lack background context when reading breaking news. This strongly validates the core concept of NewsChain.

**Q7 – Interest in Reverse-Timeline News Format (n = 12)**

| Rating | Responses | Percentage |
|---|---|---|
| 1 – Not Interested | 0 | 0% |
| 2 | 1 | 8% |
| 3 – Neutral | 2 | 17% |
| 4 | 4 | 33% |
| 5 – Very Interested | 5 | 42% |

Mean: 4.1 / 5. A total of 75% of respondents rated their interest as 4 or 5, indicating strong demand for the reverse-timeline concept that is central to NewsChain.

**Q11 – Most Wanted App Features (n = 12, multiple choice)**

| Feature | Votes | Percentage |
|---|---|---|
| Story chain visualization | 10 | 83% |
| Personalized topic feed | 9 | 75% |
| Bookmarking articles | 8 | 67% |
| Dark mode | 7 | 58% |
| Audio summary | 5 | 42% |
| Offline reading | 4 | 33% |
| Share story chain | 4 | 33% |

## 5.5 Insights from the Results

- Users frequently encounter news without sufficient context: A mean score of 3.8/5 on Q5 confirms that lack of background information is a widespread pain point, directly validating the NewsChain concept of presenting stories as historical chains.
- Strong interest in the reverse-timeline format: 75% of users rated their interest as 4 or 5, confirming the unique reverse-timeline format of NewsChain has genuine market appeal.
- Story chain visualization is the most desired feature: 83% of respondents selected it, reinforcing that the visual representation of a story's evolution is a key differentiator to build as a core feature.
- Personalization is highly valued: 75% want a personalized feed, meaning NewsChain

should include topic preference settings to tailor which story chains appear for each user.

- Users are time-constrained: Most users spend 30–60 minutes per day on news, suggesting short, scannable chain summaries are essential before users dive into full chain exploration.

# 6. Requirements Generation

## 6.1 Personas and Scenarios

### Persona 1 – Maya Chen, the Overwhelmed Student

| | |
|---|---|
| **Name** | Maya Chen |
| **Age** | 21 |
| **Occupation** | Science student at a large university |
| **Tech Use** | Heavy smartphone user; uses Instagram, Reddit, and TikTok for news |
| **Frustration** | Constantly sees breaking headlines that reference events she doesn't know about. Feels embarrassed in class discussions when she lacks context. |
| **Goal** | To quickly understand the background of a news story without hours of research |

Scenario: Maya opens her phone between classes and sees a headline: 'Parliament passes emergency AI regulation bill.' She has no idea what led to this. She opens NewsChain, taps the headline, and is immediately presented with the most recent event node. She swipes backwards through a visual chain: the earlier Senate hearings, the AI incident that triggered public debate, and the original research paper from two years ago that started it all. In under 5 minutes, Maya understands the full story. She saves the chain to review later for a class essay.

### Persona 2 – David Tanaka, the Busy Professional

| | |
|---|---|
| **Name** | David Tanaka |
| **Age** | 34 |
| **Occupation** | Product Manager at a software company |
| **Tech Use** | Uses Apple News and LinkedIn for news; checks phone during commute |
| **Frustration** | News articles are long but shallow. He wants the gist of a story's full history fast, without reading five separate articles. |
| **Goal** | Stay informed on tech and business news efficiently during his 20-minute commute |

Scenario: David is on the SkyTrain heading to work. He sees that a major tech company just announced a massive layoff. He opens NewsChain and taps the story. He is shown a concise chain: today's layoff announcement, the earnings report from last quarter, the pivot strategy announced 6 months ago, and the acquisition that started everything 2 years prior. He skims each node summary and feels fully caught up by the time he arrives at his stop. He shares the story chain link with a colleague via the app's share feature.

### Persona 3 – Sandra Kowalski, the Sceptical News Reader

| | |
|---|---|
| **Name** | Sandra Kowalski |

| | |
|---|---|
| **Age** | 40 |
| **Occupation** | High school teacher and parent of two |
| **Tech Use** | Moderate smartphone user; uses Facebook and Google News |
| **Frustration** | Distrusts sensationalist headlines; wants to see which sources are cited and whether a story has been consistent over time. |
| **Goal** | Find trustworthy, well-sourced news she can confidently discuss or share with students |

Scenario: Sandra reads a controversial headline about a new school policy being debated in her province. She opens NewsChain and navigates the chain. She notices each node displays the source publication and a source-reliability badge. She can see that the story's framing changed significantly between outlets over the months, and she can tap through to read the original government document. Feeling confident in the sourced information, she decides to use the story chain as a classroom discussion resource, having bookmarked it the night before.

**Persona 4 – Kenji Ohmura, the Casual News Scroller**

| | |
|---|---|
| **Name** | Kenji Ohmura |
| **Age** | 25 |
| **Occupation** | Barista and part-time freelance photographer |
| **Tech Use** | Heavy TikTok and Instagram user; occasionally reads news from social shares |
| **Frustration** | Finds traditional news apps boring and cluttered with too much text and unrelated ads. |
| **Goal** | Stay loosely informed on world events in a visually engaging, low-effort way |

Scenario: Kenji is on his break at work and scrolls through NewsChain's home feed. The card-based layout with bold headline images catches his eye. He swipes through a visually-presented chain about a climate summit, appreciating the clean cards with icons representing each event node. He doesn't read every card fully but skims the visual timeline enough to feel informed. He taps the 'audio summary' button and listens to a 90-second recap of the full chain while cleaning equipment, completing his news catch-up effortlessly.

## 6.2 Requirements Generated from Persona and Scenario Analysis

**From Maya Chen (Persona 1):**

- The app must present a visual reverse-timeline chain for each news story.
- Each node in the chain must include a short summary (2–3 sentences) viewable without opening the full article.
- Users must be able to bookmark or save story chains for later reference.
- The app must allow users to swipe or tap backward through a chain naturally on mobile.

**From David Tanaka (Persona 2):**

- Chain node summaries must be scannable in under 30 seconds each.
- The app must provide a 'full chain summary' view that gives an overview of the entire story arc in one scroll.
- A share feature must allow users to send a story chain link to others via messaging apps, email, etc.
- The app must function well in low-connectivity environments such as during transit.

**From Sandra Kowalski (Persona 3):**

- Each node must display the source publication name and date clearly.
- The app should include a source credibility or reliability indicator on each node.
- Users must be able to tap through to read the original full article from any node.
- Bookmarks must be organized and persistently saved across app sessions.

**From Kenji Ohmura (Persona 4):**

- The home feed must use a card-based visual layout with compelling imagery.
- The app must support a dark mode for comfortable viewing in different lighting.
- An audio summary feature should allow users to listen to a 60–90 second recap of any story chain.
- The UI must be clean and uncluttered, with no intrusive advertisements interrupting story flow.
- Chain nodes should include visual icons or category tags (e.g., Politics, Tech, Climate) for quick orientation.

# 7. Synthesis of Requirements Gathering and Generation

## 7.1 Prioritized Requirements List

### Must-Have Requirements (High Priority)

- Reverse-timeline story chain: The app must display news stories as a chain of linked events ordered from the most recent outcome backwards to the origin. This is the core differentiating feature of NewsChain.
- Node-based story display: Each event in the chain must be displayed as a distinct card/node with a headline, source, date, and a brief 2–3 sentence summary.
- Swipe-back navigation: Users must be able to navigate backwards through the chain using intuitive swipe or tap gestures, consistent with mobile UX conventions.
- Source attribution: Every node must clearly display the original news source name and publication date.
- Link to full article: Users must be able to tap any node to open the original full article, either in-app or in a browser.
- Bookmarking: Users must be able to save story chains to a personal reading list for future access.
- Home feed: A visually-engaging, card-based home screen must display trending and personalized story chains.
- Onboarding and topic selection: On first launch, users must be able to select their preferred news topics to personalize their feed.

### Want-to-Have Requirements (Medium Priority)

- Full chain summary view: A dedicated single-screen view that summarizes the entire story arc in one pass.
- Source credibility indicator: A visual badge or rating alongside each source to help users assess reliability.
- Dark mode: A full dark mode theme for the app interface to support comfortable reading at night or in low-light settings.
- Category and topic tags on nodes: Visual tags (e.g., 'Economy', 'Climate', 'Tech') to help users orient themselves within a chain.

### Low Priority / Nice-to-Have Requirements

- Offline reading: Allow users to download story chains for offline reading during low-connectivity situations.
- Push notifications: Smart, minimal notifications for major developments in a user's followed story chains.
- Search functionality: Allow users to search for specific stories, topics, or keywords to find relevant chains.
- Chain comparison: Allow users to view how different outlets frame the same chain of events side by side.
- Audio summary: Users can listen to news without using their hands.

## 7.2 Conclusions from the Requirements Process

- The core NewsChain concept of reverse-timeline news is strongly validated by both user study data and persona analysis. It directly addresses the near-universal pain point of lacking context for breaking news.
- Visual design and UX simplicity are critical: cluttered UI is the most common reason users abandon news apps. Every screen in the Figma prototype was designed to be clean and card-centric.
- Source credibility and trust are underserved in current news apps. Building transparency into each chain node is a meaningful differentiator that will attract skeptical, educated users.
- Personalization is expected by modern news readers. The onboarding topic selection and personalized home feed address this directly.

# 8. Design

## 8.1 Design Patterns Chosen

- Card-Based Layout: All news items and story chain nodes are presented as cards with a consistent structure (image, category tag, headline, metadata). This pattern supports scannability and visual hierarchy.
- Bottom Navigation Bar: Persistent navigation between four main sections (Home, Explore, Bookmarks, Profile) using icon-based tabs. Standard in mobile applications; reduces navigation complexity.
- Reverse-Timeline Chain: The core interaction pattern where users start at the latest node and progress backwards in time by tapping or swiping. This is a novel application of a timeline pattern.
- Progress Tracking: The Bookmark screen doubles as a reading progress tracker, showing percentage completion of each saved chain. This encourages re-engagement.
- Gamification: The Profile screen includes streak tracking, achievement badges, and saved/completed counts. This encourages daily news habits.

## 8.2 Rationale for Design Choices

- Yellow and dark color palette: The bold yellow accent creates high visibility and energetic branding, consistent with an app promoting active news engagement. Dark text on yellow cards ensures strong contrast and readability.
- Category tags as colored chips: Color-coded category tags (e.g., green for Environment, blue for Technology) allow users to quickly identify topic domains within a dense card list.
- Credibility score on nodes: Displaying a credibility percentage (e.g., 90% Credibility) on the ItemDetail screen addresses Sandra Kowalski's need for source trust, validated by the user study findings.
- Minimal navigation depth: All screens are accessible within one tap from the bottom navigation bar, reducing cognitive load for time-constrained users like David Tanaka.

## 8.3 Overall Workflow on Figma

The Figma prototype (available at: https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/) consists of five primary screens connected by the Bottom Navigation Bar:

- Home Screen → Explore Screen: Via the Search icon in the nav bar.
- Home Screen → Item Detail: Via tapping a news card.
- Home Screen → Bookmark Screen: Via the Bookmark icon in the nav bar.
- Home Screen → Profile Screen: Via the Profile icon in the nav bar.
- Item Detail → Chain Navigation: Users tap backwards through event nodes.

## 8.4 Screenshots of All Screens

### Screen 1: Home Feed

The Home screen displays recommended news story chains as large cards with a headline image,

category tag (e.g., Environment, Technology), trending indicator, read time, and date range showing the chain's time span. The NewsChain logo and brand name appear in the yellow header. A bottom navigation bar provides persistent access to all sections.

[Figma Screenshot – Home Screen: https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=48-33]

### Screen 2: Explore / Search

The Explore screen provides a search bar for finding news by topic, event, or keyword. Below the search bar, a horizontal scrollable row of category filter chips (All, Environment, Technology, Economy, etc.) allows quick topic filtering. A 'Trending Topics' section displays popular topic tags, followed by a list of news cards matching the current filter. This screen addresses the requirement for search functionality and topic-based discovery.

[Figma Screenshot – Explore Screen: https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=48-35]

### Screen 3: Reading Path (Bookmarks)

The Bookmark screen is branded as 'Your Reading Path' and doubles as a reading progress tracker. It displays three summary statistics: Saved chains (2), Completed chains (0), and overall Progress percentage (38%). A 'Continue Reading' section shows partially-read chains with progress bars indicating completion percentage. A 'Recommended For You' section appears below. This screen supports the requirements for persistent bookmarking and reading progress tracking.

[Figma Screenshot – Bookmark/Reading Path Screen: https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=48-66]

### Screen 4: Profile

The Profile screen shows the user's identity (username: Reader, label: Critical Thinker), a streak count (7 days), and saved/done/events statistics. A 'Recent Achievements' section displays earned badges (7 Days Streak, First Chain Completed). A 'Your Interests' section displays selected topic tags that personalize the home feed. This screen supports the gamification and personalization requirements.

[Figma Screenshot – Profile Screen: https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=48-97]

### Screen 5: Item Detail (Story Chain View)

The Item Detail screen is the core screen of NewsChain. It presents the most recent outcome node prominently at the top with a full-width hero image, category tag, headline, read time, event count, and a trending indicator. A highlighted 'The Outcome Today' section with explanatory text

introduces the reverse-timeline experience. Below it, the first chain node card is displayed showing the date, headline, and a credibility score (e.g., 90% Credibility), with a 'View Detail >' link to the full article. Bookmark and Share buttons appear in the top-right corner.

[Figma Screenshot – Item Detail Screen: https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=146-808]

# 9. Designer-Based Prototype Evaluation

## 9.1 User Journey Map

The User Journey Map below traces the primary path of a target user — represented by Persona 2 (David Tanaka, Busy Professional) — through the core NewsChain experience.

| Phase | Awareness | Exploration | Engagement | Deep Reading | Return Visit |
|---|---|---|---|---|---|
| **Touchpoint** | App store / social share | Home screen, onboarding | Story chain / Item Detail | Node tap, source view | Bookmark / Reading Path |
| **User Action** | Downloads app, sees intro | Selects topics, browses feed | Taps news card, enters chain | Swipes back through nodes, taps source | Re-opens Reading Path, continues |
| **Emotion** | Curious, hopeful | Oriented, interested | Engaged, discovering | Informed, satisfied | Habitual, accomplished |
| **Pain Point** | – | Uncertain how chain works | Chain depth unclear at first | Wants to verify credibility | Needs reminder to return |

# 10. User-Based Prototype Evaluation (User Study 2)

## 10.1 Purpose and Objectives

The objective of User Study 2 was to evaluate the usability and clarity of the NewsChain Figma prototype with real users. Specifically, the study aimed to:

- Determine whether users understand the reverse-timeline chain concept on first encounter.
- Identify any navigation pain points or confusion within the prototype flows.
- Assess the clarity of the source credibility indicator and bookmarking feature.
- Gather qualitative feedback on the visual design, card layout, and information density.
- Validate whether the prototype meets the core user needs identified in User Study 1.

## 10.2 Choice of User Study and User Sample

User Study Type: An in-person think-aloud usability test combined with a short post-task survey. Participants were asked to complete three tasks using the Figma prototype while verbalizing their thoughts.

Target Users: Same as User Study 1 — general digital news readers aged 18–45. Participants were recruited from the same social circle (friends, housemates) as Study 1.

Sample Size: 5 participants completed the prototype evaluation session.

## 10.3 User Study Tasks

- Task 1: Find a technology news story on the Home screen and open its story chain.
- Task 2: Navigate backwards through at least two event nodes in the chain.
- Task 3: Bookmark a story chain and locate it in the Reading Path screen.

## 10.4 Data Visualization and Analysis

Sample size: 5 participants. Post-task survey used a 1–5 Likert scale.

| Survey Question | Mean | Median |
|---|---|---|
| Ease of finding a story (Q1) | 4.2 / 5 | 4 |
| Clarity of reverse-timeline concept (Q2) | 3.8 / 5 | 4 |
| Ease of backward navigation (Q3) | 3.6 / 5 | 4 |
| Usefulness of credibility indicator (Q4) | 4.4 / 5 | 5 |
| Overall satisfaction with the app (Q5) | 4.0 / 5 | 4 |

Open-ended responses highlighted that the home feed was visually appealing and easy to scan. Two participants noted that the backward navigation direction was not immediately obvious and

required a short learning period. All five participants found the credibility score highly useful. One participant mentioned that an onboarding tutorial showing the chain concept would reduce initial confusion.

## 10.5 Insights from the Results

- The home feed card layout is effective: All participants found the Home screen visually engaging and easy to navigate.
- The reverse-timeline concept requires brief onboarding: 2 of 5 participants were initially unsure which direction to swipe. A short introductory animation or tooltip would resolve this.
- The credibility indicator is highly valued: Mean score of 4.4/5, confirming it is a key differentiator and should remain a Must-Have feature.
- Backward navigation affordance needs improvement: Swipe direction cues or a visible 'Previous Event' button would help users understand the chain navigation model.
- Bookmarking is intuitive: All participants successfully completed Task 3 without difficulty.

# 11. Changes to Design Based on Prototype Evaluation

## 11.1 Changes Needed Based on Feedback

- Add a brief onboarding animation or tooltip on first launch of the chain view, explaining the reverse-timeline concept and swipe direction.
- Improve backward navigation affordance by adding a visible 'Previous Event' button or directional arrow indicator below each node.
- Add a chain depth indicator (e.g., 'Event 1 of 4') so users know how many nodes are in a chain before committing to reading.
- Increase tap target size for the Bookmark button on each card to reduce accidental misses on smaller screens.
- Add a subtle animation on the chain node transition to reinforce the backward direction of travel.

## 11.2 Changes Made After Feedback

- The Reading Path screen was designed to include progress bars for saved chains, directly addressing the need for users to track their reading depth.
- Category tags were added to every news card on the home feed, Explore, and chain nodes to improve orientation.
- The credibility score badge was prominently placed on the Item Detail screen with clear percentage display.

## 11.3 Changes Not Yet Implemented (Major Revisions)

- Onboarding tutorial flow: A dedicated multi-screen onboarding experience explaining the reverse-timeline chain concept requires significant design and development work. This is deferred to the next iteration.
- Audio summary feature: A 60–90 second AI-generated audio recap of the full story chain requires backend AI integration and is beyond the current prototype scope.
- Offline reading mode: Caching full story chains for offline access requires local database storage implementation and is not feasible in the current Figma prototype stage.
- Push notifications: Smart notifications for chain updates require backend notification services and user preference management, deferred to production implementation.

## 12. Conclusion

NewsChain successfully demonstrates a novel and validated approach to mobile news consumption. By reversing the traditional chronological news feed and presenting news as visual chains of interconnected events, the app addresses a genuine and widespread user pain point: the lack of historical context when encountering breaking news headlines.

The two-phase research process — an initial user needs survey (n=12) followed by persona-driven requirements generation and prototype evaluation (n=5) — consistently validated the core product concept. 75% of initial survey respondents rated their interest in the reverse-timeline format at 4 or 5 out of 5. The Figma prototype evaluation confirmed that the home feed card layout is effective, and that the credibility indicator is highly valued by users.

The primary challenge identified is the onboarding experience: the reverse-timeline interaction model is novel and requires clear introduction for first-time users. This is the highest-priority design improvement for the next iteration.

The five screens designed in Figma — Home, Explore, Reading Path, Profile, and Item Detail — cover the full core user journey from discovery to deep reading to return visits. The implementation of the Bottom Navigation Bar and Card List components (completed in Progress Report #1) establishes the foundational UI architecture for the production build.

NewsChain has strong market differentiation from existing news apps and a clear value proposition grounded in user research. With continued iteration on the onboarding experience and chain navigation affordances, the app is positioned to deliver a genuinely transformative news reading experience.

# 13. AI Use Section

## 13.1 Table of AI Tools and Specific Use

| AI Tool Name | Version / Account Type | Specific Feature Used |
|---|---|---|
| ChatGPT | GPT-4o mini, Free account | Background research on competitor apps; example questionnaire generation; identifying key challenges in user requirements |
| Claude (Anthropic) | Claude Sonnet 4.6, claude.ai Free | Formatting Word document based on questionnaire results; Final Report writing and document structure |

## 13.2 Value Addition

All AI tools were used as assistants, not authors. The following value was added over and above what AI produced:

- All personas and scenarios were created from original observations and user study responses — AI was not used to generate personas.
- The app concept, reverse-timeline design pattern, and value proposition were independently conceived and refined.
- The questionnaire questions were drafted first by the designer; ChatGPT was then used to check clarity and suggest improvements.
- The Figma prototype — including all 5 screens, component design, color system, and interaction flows — was entirely designed by the student without AI assistance.
- All requirements were generated by the student through persona analysis; AI was not used for requirements generation.
- The implementation of UI components (Bottom Navigation Bar, Card List) in the development build was done independently by the student.

# 14. Appendix

## 14.1 Prompt History – All AI Tool Interactions

**ChatGPT (GPT-4o mini) – February 1, 2026**

Prompt: "Explain Background Research: competitor apps and inspiration apps for NewsChain."

Use: Used to compile an initial list of competitor apps (Apple News, Google News, Wikipedia, Vox) and their influence on the NewsChain concept. All content was reviewed, filtered, and rewritten by the student.

**ChatGPT (GPT-4o mini) – March 2, 2026**

Prompt 1: "Can you tell me where I should improve my questionnaire as user requirement"

Use: Reviewed the draft questionnaire for clarity improvements; student incorporated selected suggestions.

Prompt 2: "Can you tell me example of questionnaire based on the application concept"

Use: Generated example survey questions for a news application; student adapted and filtered these for relevance to NewsChain.

**ChatGPT (GPT-4o mini) – March 9, 2026**

Prompt: "What is main challenging point from user requirement"

Use: Used to identify common challenges in user requirements gathering. Output was reviewed and interpreted by the student in the context of NewsChain's specific findings.

**Claude (Anthropic) – March 9, 2026**

Prompt: "Can you format word file based on my questionnaire results?"

Use: Used to format the Requirements Document based on the student's questionnaire data and written content. All content was written by the student; Claude provided structural formatting.

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## 14.2 Figma Prototype Link

Full interactive prototype: https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/

## 14.3 Work Date / Hours Log

| Date | Hours | Description |
|---|---|---|
| Feb 1, 2026 | 6.0 | Writing summary of production proposal |
| Feb 17, 2026 | 3.0 | Making Figma app prototype |
| Feb 17, 2026 | 2.0 | Recording video, writing report |

| | | |
|---|---|---|
| Mar 2, 2026 | 1.0 | Generated requirements list from personas; organized into Must-Have / Want-to-Have categories based on Figma prototype |
| Mar 4, 2026 | 1.0 | Posting Google questionnaire and discussing app concept with housemates and friends |
| Mar 9, 2026 | 1.5 | Wrote synthesis section, conclusions, and AI use section; cross-checked requirements against Figma prototype |
| Mar 23, 2026 | 1.0 | Implemented the Bottom Navigation Bar component for the NewsChain app |
| Mar 23, 2026 | 1.0 | Implemented the Card List component for the home feed screen |
| Mar 29 2026 | 3.0 | Create multiple screens and bottom nav bar |
| Apr 10 2026 | 5.0 | Creates Remaining app functioins |
| Apr 11 2026 | 3.0 | Prototype evaluation (User Study 2); writing Final Report |
| **Total** | **27.5** | |

## 14.4 References


- Figma. (2024). Figma: The collaborative interface design tool. https://www.figma.com
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- Android Developers. (2024). Bottom Navigation. https://developer.android.com/reference/com/google/android/material/bottomnavigation/BottomNavigationView
- YouTube. (2024). Bottom Navigation Bar tutorial. https://www.youtube.com/watch?v=zI9xXutdZb0